



NITI Aayog



SECTOR AT A GLANCE 06

Urban: Household Water Supply in India

WASH in India

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SECTOR AT A GLANCE 06

URBAN: HOUSEHOLD WATER SUPPLY

WASH IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Urban: Household Water Supply. NITI Aayog Sector-at-a-Glance Series – WASH in India, 2023-WASH-SAG-06, <link>

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Overview

Ensuring household-level water supply in urban areas has been a key focus area for the Government over the last few years. Significant funds have been allocated towards this, and programs such as AMRUT 2.0 and several State initiatives have worked enabling universal tap water access to all urban households.

Such a mission-mode focus supported by significant funding allocation has led to significant progress in coverage. The number of urban households that reported access to a principal source of drinking water that was for the exclusive use of households has shown a marked improvement in 2021 compared to 2012¹ increasing from 47% to 65%.

India is now well positioned to further improve urban water supply through an emphasis on source sustainability, ensuring potable water quality, reduction of non-revenue water, maintenance of water supply infrastructure, the utilization of hybrid models of water delivery, and engaging with citizens with a focus on water quality.

Specific themes that states and cities could focus on include investments in IOT and analytics to improve water grid performance, partnerships with decentralized water delivery models to ensure universal coverage in the city, setting up water testing facilities, especially in smaller cities and towns, and finally investing in IEC and behaviour change campaigns to ensure citizens conserve and treat water if needed, before consumption.

Figure 1: Key urban household water supply statistics

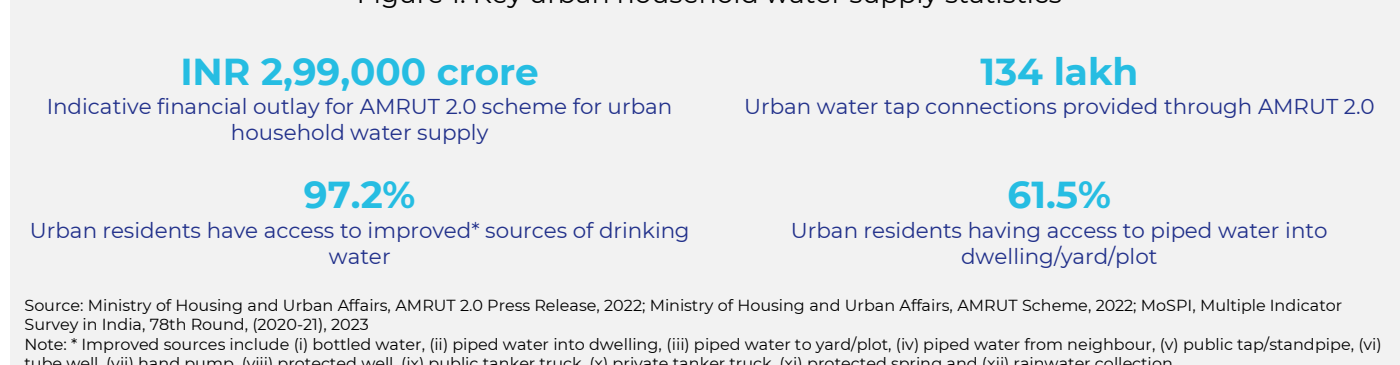
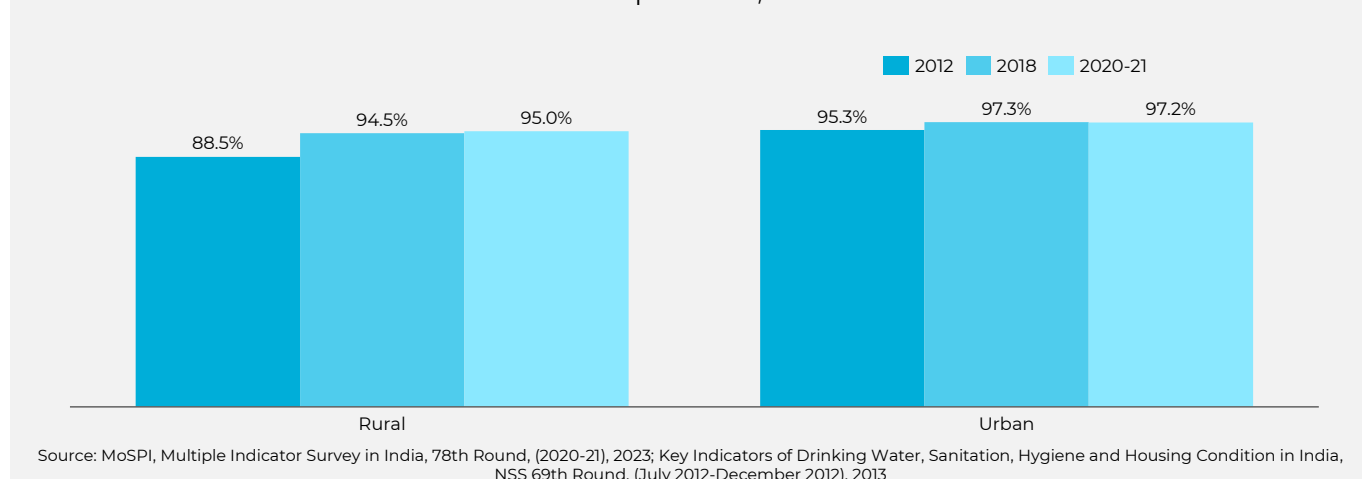


Figure 2: Percentage of persons reported to have improved source of drinking water located in the household premises, 2012-2021



¹ MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023

Key trends

This section dives deeper into India's urban household water supply landscape, outlining statistics and trends on:

1. Water access
2. Water quality and treatment

Water access

The provision of piped water in rapidly urbanising cities in India is a key policy priority. The AMRUT 2.0 scheme being implemented by the Ministry of Housing and Urban Affairs targets to bridge the gaps in urban water supply by providing functional tap water connections to all the urban households in the country.

Access to improved sources of drinking water has improved in urban India:

76% of urban residents report having access to an improved source of drinking water located within household premises that was sufficient for meeting needs in 2021, compared to 72% in 2018.² Improved sources of water include bottled water, piped water into the dwelling, yard or plot, or from a neighbour; water from public tap, tube well, hand pump, protected well, public and private tanker, protected spring, and rainwater collection.³

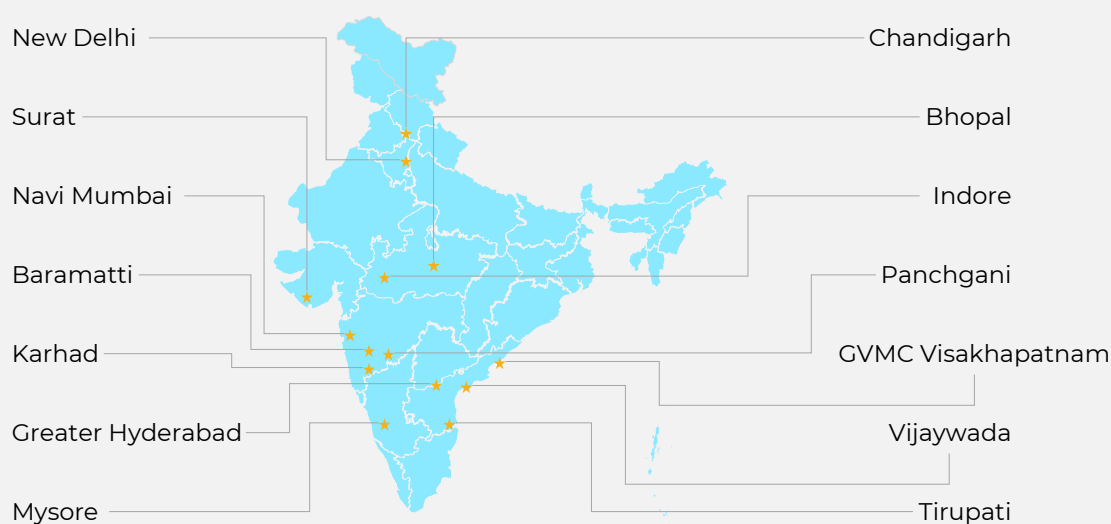
Access to principal sources of drinking water for the exclusive use of households has also increased:

The number of urban households who reported access to a principal source of drinking water that was for the exclusive use of households stood at ~65% in 2021, a marked improvement from ~47% in 2012.⁴

Access also varies across income levels:

Access to piped water supply is 45% among the lowest expenditure quintiles compared to 69% among the highest expenditure quintiles.⁵

Figure 3: Water+ certified cities in India



Source: PIB, Treatment of Waste Water, 2023

Note: The Water+ certification evaluates whether cities and urban local bodies are collecting and safely transporting and treating used water or faecal sludge and ensuring maximum reuse after secondary or tertiary treatment.

2 MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023; MoSPI, Drinking Water, Sanitation, Hygiene and Housing Condition in India, NSS 76th Round, (July 2018-December 2018), 2019

3 MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023; MoSPI, Drinking Water, Sanitation, Hygiene and Housing Condition in India, NSS 76th Round, (July 2018-December 2018), 2019

4 MoSPI, Multiple Indicator Survey in India, 78th Round, (2020-21), 2023

5 International Institute for Population Sciences (IIPS) and ICF, National Family Health Survey (NFHS-5), 2019-21: India: Volume I, 2022

Water quality and treatment

Water quality and treatment in urban areas for household water supply involve monitoring, purification, and maintenance of infrastructure to ensure safe and clean drinking water.

Quality of urban water supply can be improved:

A 2019 survey by the BIS on the quality of water in 21 cities found that most of the samples from all cities except Mumbai failed to meet the standards prescribed by BIS.⁶

Water supply sources are polluted:

Cities in India rely on both groundwater as well as surface water for urban water supply. Groundwater contamination has been reported in approximately one in ten groundwater assessment units with harmful chemicals such as fluoride and arsenic found beyond the permissible limits.⁷ At the same time, the level of pollution in surface water has also become a serious concern as 351 river stretches along 323 rivers were found polluted in 2018.⁸

⁶Ministry of Consumer Affairs, Food & Public Distribution, Report of Testing of Piped Drinking Water Samples from 20 Capital Cities and Delhi, 2019

⁷ Government of India, Integrated government online directory – State-wise list of districts; Ministry of Jal Shakti, PIB, Contamination of Ground Water Due to Arsenic and Fluoride, 2021; Lok Sabha Secretariat, Standing Committee on Water Resources, 2023

⁸ CPCB, Polluted River Stretches for Restoration of Water Quality, 2022

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective urban household water supply practices. These revolve primarily around the potential for better health outcomes, improved gender equity and improved water-use efficiency.



Improved health outcomes:

Ensuring potable water supply in all households can drive health outcomes, reduce absenteeism from schools and work, lower health expenditure, driving further socio-economic benefits.⁹ For instance, research shows that regulated tap water supply can reduce the risk of exposure to natural contaminants such as arsenic and fluoride or anthropogenic such as nitrate, contributing to around 5% of the total disease burden in the country.¹⁰



Improved gender equity and productivity for women:

Access to piped water reduces women's time burden and drudgery related to procuring water from community taps or distant locations, thus alleviating time poverty challenges. Women can participate more regularly in education and other productive market activities by reducing the time and labour needed for water collection.¹¹



Improved water supply efficiency and sustainability by reducing Non-Revenue Water (NRW):

NRW is defined as the difference between the volume of water introduced into the water distribution system and the volume billed to end-users.¹² Research indicates that by reducing NRW, India can achieve significant water savings, drive sustainability of utilities, and enhance water use efficiency. Further, approximately 600 million people experience extremely high water stress, resulting in health challenges and even deaths due to insufficient access to safe water.¹³ With reduced water losses through leak detection and better infrastructure, there is a potential to meet increasing water demands, ensure equitable water distribution, ensure viable distribution models, and save lives.

Challenges

This section highlights some of the major challenges and constraints facing urban household water supply in India. Efforts to resolve such challenges will ensure better water access, improved water supply infrastructure and overall health and gender outcomes.



Inadequate pricing mechanisms and high NRW levels:

In India, the average NRW is 38%, higher than the global average and the norm of 20% set by the MoHUA.¹⁴ This is significant considering the severe water stress faced by major Indian cities.¹⁵ Further, a significant proportion of the water supplied to Indian cities does not earn any revenue.¹⁶ Some projects piloted under the design-finance-build-operate-transfer model received mixed responses primarily due to low willingness to pay among urban slum residents.¹⁷ Addressing this will be critical going forward.

9 WHO, INDIA: Tap water at home lowers disease, saves time and labour, 2021

10 WHO, INDIA: Tap water at home lowers disease, saves time and labour, 2021

11 ADB, Impact of Infrastructure on Women's Time Poverty in Asia and the Pacific, 2015

12 Ministry of Housing and Urban Affairs, Extent of Non-Revenue Water, 2021

13 Ministry of Jal Shakti, Deaths Due to Lack of Clean Water, 2022; NITI Aayog, Composite Water Management Index 1.0, 2018

14 Ministry of Housing and Urban Affairs, Extent of Non-Revenue Water, 2021

15 WWF, Cities across the globe face an alarming rise in water risks, 2020

16 UNESCO, National Institute of Urban Affairs, A Qualitative Framework to Evaluate the Extent of Integrated Urban Water Management in Indian Cities, 2021

17 NITI Aayog, Composite Water Resources Management: Performance of States, 2019



Badly maintained water supply infrastructure that drives water contamination:

Research finds that even when the water quality supplied by municipal plants is adequate, it deteriorates as it travels through insufficiently maintained last-mile pipelines that are often laid near stormwater drains or municipal sewers.¹⁸



Depletion of groundwater and climate change likely to intensify drinking water scarcity in the future:

It is predicted that 30 Indian cities will face a significant risk of running out of water by 2050.¹⁹ Falling groundwater levels are a major risk factor since groundwater accounts for 50% of all urban water supply.²⁰



Limited availability of water quality-related data:

Most Indian cities do not have well-maintained and granular databases on water quality and supply aspects. Individual agencies collect data as per their needs and requirements, which are often not shared with the public or each other.²¹ Lack of standardised data and public access on key indicators of water supply challenges monitoring, comparison, or evidence-based decision-making.²²

¹⁸ CGWB, Dynamic Ground Water Resources of India, 2022

¹⁹ WWF, Cities across the globe face an alarming rise in water risks, 2020

²⁰ Central Ground Water Board (CWGB), Dynamic Ground Water Resources of India, 2022

²¹ UNESCO, A Qualitative Framework to Evaluate the Extent of Integrated Urban Water Management in Indian Cities & Applying the Framework to Delhi, 2021

²² UNESCO, A Qualitative Framework to Evaluate the Extent of Integrated Urban Water Management in Indian Cities & Applying the Framework to Delhi, 2021

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee urban household water supply and related practices.



Ministry of Housing & Urban Affairs²³

The ministry oversees and manages the supply of drinking water, rejuvenation of water bodies, waste management, and rainwater harvesting in urban areas. It is responsible for the implementation of the Atal Mission for Rejuvenation and Urban Transformation 2.0. The ministry also houses the Central Public Health and Environmental Engineering Organisation which provides technical inputs in policy formulation to the MoHUA on matters related to urban water supply and supports the states by providing technical advice, and guidelines and propagating new technologies in the field.



Ministry of Jal Shakti²⁴

The Ministry of Jal Shakti is responsible for national water planning and coordination for the development, conservation, and management of water as a national resource. It coordinates with the MoHUA on matters of urban water supply.



Central Pollution Control Board²⁵

The Board is responsible for the prevention, control, and abatement of water pollution (thus including pollution caused by industries) in accordance with the Water (Prevention and Control of Pollution) Act, 1974.

Major policies and programs

This section highlights existing policies related to urban household water supply and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI for States platform.



National Water Policy²⁶:

It aims to manage India's scarce water resources and ensure sustainable use of water. It consists of provisions that regulate the domestic supply of water in urban areas by outlining the sources of domestic water supply in urban areas, laying out rules for water treatment systems and promoting data collection initiatives.



Atal Mission for Rejuvenation and Urban Transformation 2.0 (AMRUT 2.0)²⁷

AMRUT 2.0 aims to provide tap water connections to all urban households through several complementary measures:

- Mandating PPP models in million-plus cities
- Supporting innovative, low-cost indigenous water treatment technologies
- Incentivising the successful implementation of 24x7 'Drink from Tap' projects

Ensuring community participation in the management of water infrastructure and quality assessment

²³ Ministry of Housing and Urban Affairs, Mandate, Accessed December 2023

²⁴ Ministry of Jal Shakti, Department of Drinking Water and Sanitation, Accessed December 2023; Ministry of Housing and Urban Affairs, Mandate, Accessed December 2023

²⁵ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

²⁶ Ministry of Jal Shakti, National Water Policy, 2012

²⁷ MoHUA, AMRUT 2.0: Operational Guidelines, 2021

Way forward

India has made a significant commitment to ensuring household-level improved water supply to all its urban citizens. This has the potential to enable significant benefits in terms of water availability, improved gender equity and productivity for women, and improved health outcomes.

However, going forward care must be taken to ensure that the water being supplied is potable, that water supply infrastructure is well maintained, that ULBs are able to run their operations sustainably, that water sources, especially groundwater, are well managed, and that equity considerations are kept in mind to ensure that all segments benefit from these programs.

Some focus areas that could spur progress include:

 Adopting innovative technologies for urban water management	Technologies such as IOT, data analytics, and AI-enabled systems can improve the efficiency of the urban water supply grids. This would enable preventive maintenance, leak detection, water quality detection, and improvements to NRW levels. India can learn from a range of global examples and implement this at scale.
 Advanced hybrid approaches for water coverage	Hybrid approaches that combine centralized water grids and also decentralized water supply through SMEs and water kiosk operators can enhance urban water supply. Several cities in India, and across the world, are increasingly adopting a hybrid approach that combines these systems to ensure universal coverage and constant operations. Indian cities should partner with organizations and financial institutions that are promoting these models to drive water access at scale.
 Enhance the density of water quality monitoring	States and cities should invest in collecting more granular data related to water quality across different parts of cities and then planning remedial measures. Partnerships with private sector operators to improve the capacity of water quality testing labs can be used for this.
 Invest in behaviour change programs focused on water quality	A large fraction of water users are not fully aware of the need to adequately treat water before consumption. They will likely get exposed to unpotable water in case there is disruption or leaks in the supply infrastructure. They would benefit from behaviour change programs and IEC campaigns focused on the need to adequately treat the water supply that is being received at the household level.

Recommended readings

UNESCO. National Institute of Urban Affairs. (2020). *White paper: a qualitative framework to evaluate the extent of integrated urban water management in Indian cities and applying the framework to Delhi.*

<https://unesdoc.unesco.org/ark:/48223/pf0000375500>

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<https://www.who.int/india/news/feature-stories/detail/tap-water-at-home-lowers-disease-saves-time-and-labour>

USAID. TERI (2017). *State of Urban Water and Sanitation in India.*

<https://www.terisas.ac.in/pdf/StateofWASHReport.pdf>

Ministry of Jal Shakti (2023). *Groundwater: A Valuable but Diminishing Resource.*

https://loksabhadocs.nic.in/lsscommittee/Water%20Resources/17_Water_Resources_22.pdf



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